B.TECH · 1ST YEAR · 1ST SEMESTER

Engineering Mathematics

Complete Formula & Theorem Revision Sheet

📐 UNIT III — CALCULUS

Mean Value Theorems & Series Expansions

📋 CONTENTS AT A GLANCE

EXAM-READY · FORMULA-FOCUSED · LAST-DAY REVISION TOOL

COURSE OBJECTIVES — UNIT III

Objective 1 — Behaviour of Functions

- ✓ Understand how derivatives describe rate of change
- ✓ Use Mean Value Theorems to analyze function behaviour on intervals
- ✓ Locate points where instantaneous rate = average rate
- ✓ Identify horizontal tangents using Rolle's Theorem

Objective 2 — Derivative Applications

- ✓ Apply LMVT to prove inequalities and estimate values
- ✓ Use CMVT for comparing two functions simultaneously
- ✓ Prove existence results using intermediate theorems
- ✓ Connect derivatives to geometric slope interpretation

Objective 3 — Taylor & Maclaurin Series

- ✓ Expand functions as infinite polynomial sums
- ✓ Compute approximations to any desired accuracy
- ✓ Use series to simplify limits and complex integrals
- ✓ Identify when and where a series converges

FUNDAMENTAL CONDITIONS FOR ALL THEOREMS

CONTINUITY — What It Means

$$\lim_{x\to c} f(x) = f(c) \text{ for all } c \in [a, b]$$

- ✓ No jumps, holes, or breaks in the graph
- ✓ Left-hand limit = Right-hand limit = Function value
- ✓ Required on the **closed** interval [a, b]
- → Why needed: Ensures no sudden gaps exist; theorem logic requires the function to be "traceable" continuously

DIFFERENTIABILITY — What It Means

$$f'(x) = \lim_{h\to 0} [f(x+h)-f(x)] / h \text{ exists}$$

- ✓ Tangent (derivative) exists at every point
- ✓ No sharp corners, cusps, or vertical tangents
- ✓ Required on the **open** interval (a, b)
- → Why needed: Without a derivative, we cannot guarantee a point c where $f'(c)$ equals the slope expression

Key Distinction: Continuity is checked on **[a, b]** (closed), Differentiability on **(a, b)** (open). Differentiability ⟹ Continuity, but NOT vice versa!

CONDITIONS SUMMARY TABLE

Condition	Interval	Mathematical Form	Why Required	Fails When
Continuity	[a, b] (closed)	$\lim f(x) = f(c)\ \forall c\in[a,b]$	No "gaps" in domain	Jump/removable discontinuity
Differentiability	(a, b) (open)	$f'(x)$ exists $\forall x\in(a,b)$	Tangent must exist for the conclusion	Sharp corner, $\|x\|$ at x=0
Equal Endpoints	At a and b	$f(a) = f(b)$	Rolle's only — ensures return to same height	$f(a) \neq f(b)$

STATEMENT OF ROLLE'S THEOREM

ROLLE'S THEOREM — Formal Statement

If a function **f(x)** satisfies **all three conditions**:

CONDITION 1	CONDITION 2	CONDITION 3
f is continuous on [a, b]	f is differentiable on (a, b)	f(a) = f(b)

⟹ Then there exists at least one c ∈ (a, b) such that:

$$f'(c) = 0$$

GEOMETRICAL INTERPRETATION

Visual Meaning

- ✓ Graph starts and ends at the **same height**
- ✓ At point c, the tangent is **perfectly horizontal**
- ✓ Like a ball thrown upward — must have zero velocity at peak
- ✓ The curve must "turn around" somewhere in (a, b)

◆ **Horizontal Tangent:** $f'(c) = 0$ means the slope of the tangent line at $x = c$ is **zero** (tangent is parallel to x-axis)

Physical Meaning

- ✓ If a car starts and ends at the same position, its velocity must be zero at some instant
- ✓ A ball thrown up returns to same height → velocity = 0 at highest point
- → "At least one" c → could be multiple such points

Special Note: c need not be unique. There may be **multiple** points where $f'(c) = 0$

KEY PROPERTIES & EDGE CASES

Scenario	What Happens	Example Function
All 3 conditions met	Guaranteed: ∃ c ∈ (a,b) with $f'(c) = 0$	$f(x) = x^2 - x$ on [0,1]
$f(a) \neq f(b)$	Rolle's theorem does NOT apply	$f(x) = x$ on [0, 1]
Not differentiable at a corner	Rolle's theorem may FAIL	$f(x) = \|x\|$ on [-1, 1]
Constant function	Every point satisfies $f'(c) = 0$	$f(x) = k$ (constant)

HOW TO APPLY ROLLE'S THEOREM (STEPS)

Step-by-Step Verification

①	②	③	④
Check Continuity on [a, b]	Check Differentiability on (a, b)	Verify f(a) = f(b)	Solve f'(x) = 0 → find c

STATEMENT OF LMVT

LAGRANGE'S MEAN VALUE THEOREM — Formal Statement

If **f(x)** satisfies **two conditions**:

CONDITION 1	CONDITION 2
f is continuous on [a, b]	f is differentiable on (a, b)

⟹ Then there exists at least one $c \in (a, b)$ such that:

$$f'(c) = [f(b) - f(a)] / (b - a)$$

Equivalently: $f(b) - f(a) = (b - a) \cdot f'(c)$

INTERPRETATION & GEOMETRIC MEANING

Geometric Meaning

- ✓ **[f(b) − f(a)] / (b − a)** = slope of secant line joining (a, f(a)) to (b, f(b))
- ✓ **f′(c)** = slope of tangent line at point c
- ✓ LMVT says: *At some point c, the tangent is parallel to the secant*
- ✓ The tangent slope equals the chord slope at point c

Physical Interpretation

- ✓ **Average rate of change** (secant) over [a, b] equals **instantaneous rate** at some c
- ✓ A car travelling 100 km in 1 hour must have speedometer showing exactly 100 km/h at some moment
- → LMVT is a generalization of Rolle's Theorem (Rolle's is a special case when f(a) = f(b))

RELATIONSHIP: ROLLE'S ↔ LMVT

Rolle's Theorem is a Special Case of LMVT:
If f(a) = f(b), then LMVT gives: $f'(c) = [f(b) - f(a)] / (b - a) = 0/(b - a) = 0$
⟹ $f'(c) = 0$, which is exactly Rolle's conclusion!

ALTERNATIVE FORMS OF LMVT

STANDARD FORM	INCREMENT FORM	WITH H = B − A
$f'(c) = [f(b)-f(a)] / (b-a)$	$f(b) = f(a) + (b-a)\cdot f'(c)$	$f(a+h) = f(a) + h\cdot f'(a+\theta h)$ where $0 < \theta < 1$

APPLICATIONS OF LMVT

Application Type	How LMVT is Used	Example Statement
Proving Inequalities	Apply LMVT, then bound f′(c)	Prove: $\lvert\sin a - \sin b\rvert \le \lvert a - b\rvert$
Estimating Function Values	Use known f(a) and f′(c) range	Estimate $\sqrt{4.1}$ using LMVT on $\sqrt{x}$
Existence of Roots	Show f′(c) = k has a solution	Prove f′(c) = 2 for some $c \in (0,2)$
Monotonicity Test	If f′ > 0 everywhere → f is increasing	f is strictly increasing on (a,b)

STEPS TO VERIFY LMVT

Step-by-Step Application

① Verify	② Verify	③ Compute	④ Find f′(x)	⑤ Solve
Continuity on [a,b]	Differentiability on (a,b)	[f(b)−f(a)]/(b−a)	Differentiate f(x)	f′(c) = [f(b)−f(a)]/(b−a)

Cauchy's Mean Value Theorem (CMVT)

STATEMENT OF CMVT

CAUCHY'S MEAN VALUE THEOREM — Formal Statement

If **f(x)** and **g(x)** both satisfy on [a, b]:

CONDITION 1	CONDITION 2	CONDITION 3
Both continuous on [a, b]	Both differentiable on (a, b)	$g'(x) \neq 0$ for all $x \in (a,b)$

⟹ Then there exists at least one $c \in (a, b)$ such that:

$$f'(c) / g'(c) = [f(b) - f(a)] / [g(b) - g(a)]$$

⚠ *Note: $g(b) \neq g(a)$ is guaranteed by $g'(x) \neq 0$ on (a,b) [if $g' = 0$ somewhere, Rolle's would give $g(a)=g(b)$]*

INTERPRETATION OF CMVT

Ratio Interpretation

- ✓ LHS: Ratio of instantaneous rates of f and g at point c
- ✓ RHS: Ratio of total changes of f and g over [a, b]
- ✓ CMVT connects ratio of derivatives to ratio of differences
- → Useful when two functions are related parametrically (like x = g(t), y = f(t))

Parametric Interpretation

- ✓ Think of a curve parametrized as: $(g(t), f(t))$
- ✓ $f'(c)/g'(c)$ = slope of tangent at parameter c
- ✓ $[f(b)-f(a)]/[g(b)-g(a)]$ = slope of chord
- ✓ CMVT: tangent slope = chord slope at some c

CMVT → LMVT (SPECIAL CASE)

Set g(x) = x: Then $g'(x) = 1$, $g(b) - g(a) = b - a$
CMVT becomes: $f'(c) / 1 = [f(b) - f(a)] / (b - a)$ ⟹ **LMVT!**
So LMVT is a special case of CMVT when g(x) = x.

KEY CONDITIONS — DO NOT MISS THESE!

⚠ **Critical:** $g'(x) \neq 0$ on (a, b) must be explicitly verified — else the denominator $g(b) - g(a)$ could be zero!

💡 **Remember:** Both functions must be continuous on [a, b] AND differentiable on (a, b). If only one fails, CMVT cannot be applied.

CMVT VS LMVT — DIRECT COMPARISON

Feature	LMVT	CMVT
Functions Involved	One function: f(x)	Two functions: f(x) and g(x)
Formula	$f'(c) = [f(b)-f(a)]/(b-a)$	$f'(c)/g'(c) = [f(b)-f(a)]/[g(b)-g(a)]$
Extra Condition	None	$g'(x) \neq 0$ on (a, b)
Special Case	LMVT is CMVT with g(x)=x	CMVT generalizes LMVT
Used For	Single function analysis	Comparing two functions, L'Hôpital proof

CMVT is used to PROVE L'Hôpital's Rule for 0/0 and ∞/∞ forms

📊 MASTER COMPARISON TABLE — ALL THREE THEOREMS

Feature	Rolle's Theorem	LMVT (Lagrange)	CMVT (Cauchy)
Full Name	Rolle's Theorem	Lagrange's Mean Value Theorem	Cauchy's Mean Value Theorem
Functions	1 function: $f(x)$	1 function: $f(x)$	2 functions: $f(x)$, $g(x)$
Continuity	f on $[a, b]$	f on $[a, b]$	f and g on $[a, b]$
Differentiability	f on (a, b)	f on (a, b)	f and g on (a, b)
Extra Condition	$f(a) = f(b)$	None	$g'(x) \neq 0$ on (a,b)
Conclusion	$\exists\ c: f'(c) = 0$	$\exists\ c: f'(c) = [f(b)-f(a)]/(b-a)$	$\exists\ c: f'(c)/g'(c) = [f(b)-f(a)]/[g(b)-g(a)]$
Geometric Meaning	Horizontal tangent exists	Tangent ∥ to secant chord	Tangent ratio = chord ratio (parametric)
Physical Meaning	Velocity = 0 at some instant	Instantaneous speed = Average speed	Ratio of velocities = Ratio of avg. changes
Special Case of	LMVT (when $f(a)=f(b)$)	CMVT (when $g(x) = x$)	Most General Theorem
Application	Finding roots of $f'(x) = 0$	Inequalities, approximations	Proving L'Hôpital's Rule

🔗 HIERARCHY DIAGRAM

CAUCHY'S MVT (Most General)

↓ Set $g(x) = x$

LAGRANGE'S MVT (LMVT)

↓ Set $f(a) = f(b)$

ROLLE'S THEOREM (Special Case)

⚡ QUICK FORMULA REFERENCE

ROLLE'S

$$f'(c) = 0$$

$c \in (a, b)$ where $f(a) = f(b)$

LMVT

$$f'(c) = [f(b)-f(a)] / (b-a)$$

Slope of tangent = slope of chord

CMVT

$$f'(c)/g'(c) = [f(b)-f(a)]/[g(b)-g(a)]$$

Generalization for two functions

TAYLOR'S THEOREM — STATEMENT

TAYLOR'S THEOREM (General Form)

If f(x) has (n+1) continuous derivatives on [a, b], then for any $x \in [a, b]$:

$$f(x) = f(a) + (x-a)\cdot f'(a) + (x-a)^2/2! \cdot f''(a) + (x-a)^3/3! \cdot f'''(a) + \cdots + (x-a)^n/n! \cdot f^n(a) + R_n(x)$$

$$f(x) = \Sigma\ [k=0 \text{ to } n]\ (x-a)^k / k! \cdot f^{(k)}(a) + R_n(x)$$

REMAINDER TERM (LAGRANGE FORM)

Lagrange's Remainder

$$R_n(x) = (x-a)^{n+1} / (n+1)! \cdot f^{(n+1)}(c)$$

where c is some point between a and x (i.e., $c \in (a, x)$)

- ✓ Gives the exact error in truncating the series
- ✓ If $R_n(x) \to 0$ as $n \to \infty$, the series converges to f(x)

Cauchy's Remainder

$$R_n(x) = (x-c)^n(x-a) / n! \cdot f^{(n+1)}(c)$$

where $0 < \theta < 1$, $c = a + \theta(x - a)$

- ✓ Alternative form — same existence guarantee
- ✓ Less commonly used in B.Tech problems

TAYLOR SERIES (INFINITE FORM)

TAYLOR SERIES — When $R_n(x) \to 0$

If $R_n(x) \to 0$ as $n \to \infty$, the function equals its Taylor series:

$$f(x) = f(a) + f'(a)(x-a) + f''(a)(x-a)^2/2! + f'''(a)(x-a)^3/3! + \cdots + f^n(a)(x-a)^n/n! + \cdots \text{ (to } \infty)$$

$$f(x) = \Sigma\ [n=0 \text{ to } \infty]\ f^{(n)}(a) \cdot (x-a)^n / n!$$

TAYLOR WITH SUBSTITUTION: X = A + H

$$f(a + h) = f(a) + h\cdot f'(a) + h^2/2!\cdot f''(a) + h^3/3!\cdot f'''(a) + \cdots$$

Here h = x − a. This form is commonly used in numerical methods.

KEY POINTS ABOUT TAYLOR'S THEOREM

Property	Detail
Convergence	Series converges if $R_n(x) \to 0$ as $n \to \infty$ within the interval of convergence
Approximation	Truncating to n terms gives a polynomial approximation with error $R_n(x)$
Required	f must be infinitely differentiable (or at least n+1 times) near x = a
Centre of Expansion	a is the "centre" or "base point" around which expansion is done
Maclaurin vs Taylor	Maclaurin = Taylor with a = 0 (expansion about origin)

MACLAURIN SERIES — DEFINITION

MACLAURIN SERIES — Taylor Series at a = 0

$$f(x) = f(0) + x \cdot f'(0) + x^2/2! \cdot f''(0) + x^3/3! \cdot f'''(0) + \cdots$$

$$f(x) = \Sigma\ [n=0 \text{ to } \infty]\ f^{(n)}(0) \cdot x^n / n!$$

Special case of Taylor's series with centre a = 0.

STANDARD MACLAURIN EXPANSIONS

① e^x — Exponential Function | Valid for all $x \in (-\infty, \infty)$

$$e^x = 1 + x + x^2/2! + x^3/3! + x^4/4! + \cdots = \Sigma\ x^n/n!$$

All derivatives at 0: $f^{(n)}(0) = 1$ for all n. $e^1 = 1 + 1 + 1/2! + 1/3! + \cdots \approx 2.718$

② sin x — Sine Function | Valid for all $x \in (-\infty, \infty)$

$$\sin x = x - x^3/3! + x^5/5! - x^7/7! + \cdots = \Sigma\ (-1)^n\ x^{2n+1}/(2n+1)!$$

Only ODD powers of x. Signs alternate: +, −, +, −, ⋯

③ cos x — Cosine Function | Valid for all $x \in (-\infty, \infty)$

$$\cos x = 1 - x^2/2! + x^4/4! - x^6/6! + \cdots = \Sigma\ (-1)^n\ x^{2n}/(2n)!$$

Only EVEN powers of x. Starts with 1. Signs alternate: +, −, +, −, ⋯

④ log(1+x) = ln(1+x) | Valid for $-1 < x \le 1$

$$\ln(1+x) = x - x^2/2 + x^3/3 - x^4/4 + \cdots = \Sigma\ (-1)^{n+1}\ x^n/n$$

No factorials in denominator! Coefficients are 1/n. Converges only for $-1 < x \le 1$.

⑤ $(1+x)^n$ — Binomial Series | Valid for $|x| < 1$ (when n is not a positive integer)

$$(1+x)^n = 1 + nx + n(n-1)/2! \cdot x^2 + n(n-1)(n-2)/3! \cdot x^3 + \cdots$$

For positive integer n: finite series (binomial theorem). For other n: infinite series valid for $|x| < 1$.

STANDARD EXPANSIONS SUMMARY TABLE

Function	First 4 Terms	General Term	Valid for
e^x	$1 + x + x^2/2! + x^3/3!$	$x^n/n!$	All x
sin x	$x - x^3/3! + x^5/5! - x^7/7!$	$(-1)^n\ x^{2n+1}/(2n+1)!$	All x
cos x	$1 - x^2/2! + x^4/4! - x^6/6!$	$(-1)^n\ x^{2n}/(2n)!$	All x
ln(1+x)	$x - x^2/2 + x^3/3 - x^4/4$	$(-1)^{n+1}\ x^n/n$	$-1 < x \le 1$
$(1+x)^n$	$1 + nx + n(n-1)x^2/2! + \cdots$	$C(n,k)\ x^k$ for $k=0,1,2,\cdots$	$|x| < 1$

Memory Trick: e^x → all powers, both even & odd | sin x → odd powers, starts with x | cos x → even powers, starts with 1 | ln(1+x) → all powers, no factorials!

Applications of Taylor & Maclaurin Series

WHEN TO USE TAYLOR VS MACLAURIN

Use TAYLOR Series when...

- ✓ Expanding about a point **$a \neq 0$**
- ✓ Need local approximation near $x = a$
- ✓ Computing $f(x)$ for x close to a (not close to 0)
- ✓ Example: Expand $\sin(x)$ near $x = \pi/2$
- ✓ Used in: Engineering approximations, error analysis

Use MACLAURIN Series when...

- ✓ Expanding about **$a = 0$** (origin)
- ✓ x is small: $x \approx 0$ or $|x| << 1$
- ✓ Using standard functions: e^x, $\sin x$, $\cos x$, $\ln(1+x)$
- ✓ Example: Approximate $\sin(0.1)$ or $e^{0.2}$
- ✓ Used in: Small angle approximation, physics problems

APPROXIMATION OF VALUES USING SERIES

Value to Find	Series to Use	Approximation Formula	Result
$e \approx ?$	e^x at $x=1$	$1 + 1 + 1/2! + 1/3! + 1/4! + \cdots$	≈ 2.71828
$\sin(0.1) \approx ?$	$\sin x$ at $x=0.1$	$0.1 - (0.1)^3/6 = 0.1 - 0.000167$	≈ 0.09983
$\cos(0.1) \approx ?$	$\cos x$ at $x=0.1$	$1 - (0.1)^2/2 = 1 - 0.005$	≈ 0.99500
$\ln(1.1) \approx ?$	$\ln(1+x)$ at $x=0.1$	$0.1 - (0.1)^2/2 + (0.1)^3/3$	≈ 0.09531
$e^{0.1} \approx ?$	e^x at $x=0.1$	$1 + 0.1 + 0.01/2 + \cdots$	≈ 1.10517

SMALL ANGLE APPROXIMATIONS (IMPORTANT!)

FOR SMALL X (RAD)
$\sin x \approx x$

FOR SMALL X (RAD)
$\cos x \approx 1 - x^2/2$

FOR SMALL X (RAD)
$\tan x \approx x$

SIMPLIFYING LIMITS USING SERIES

Series for Limit Evaluation

Method: Replace function with its series expansion, then simplify

EXAMPLE 1:

$$\lim_{x\to 0} \sin(x)/x = \lim_{x\to 0} (x - x^3/6 + \cdots)/x = \lim_{x\to 0} (1 - x^2/6 + \cdots) = 1$$

EXAMPLE 2:

$$\lim_{x\to 0} (e^x - 1)/x = \lim_{x\to 0} (x + x^2/2 + \cdots)/x = \lim_{x\to 0} (1 + x/2 + \cdots) = 1$$

DERIVED SERIES FROM STANDARD EXPANSIONS

Derived Series	How Obtained	Expansion
e^{-x}	Replace x by $-x$ in e^x	$1 - x + x^2/2! - x^3/3! + \cdots$
$\sin^2 x$	Use identity: $\sin^2 x = (1-\cos 2x)/2$	$x^2 - x^4/3 + 2x^6/45 - \cdots$
$\ln(1-x)$	Replace x by $-x$ in $\ln(1+x)$	$-x - x^2/2 - x^3/3 - x^4/4 - \cdots$
$\sinh x$	$(e^x - e^{-x})/2$	$x + x^3/3! + x^5/5! + \cdots$
$\cosh x$	$(e^x + e^{-x})/2$	$1 + x^2/2! + x^4/4! + \cdots$

QUICK MEMORY CARDS — SCAN & RECALL

Q1: 3 conditions for Rolle's Theorem?

A: (1) f continuous on $[a,b]$ (2) f differentiable on (a,b) (3) $f(a) = f(b)$

Q2: Conclusion of Rolle's Theorem?

A: $\exists\, c \in (a,b)$ such that $f'(c) = 0$

Q3: Geometric meaning of Rolle's Theorem?

A: There exists a horizontal tangent line (slope = 0) at some point c in (a,b)

Q4: LMVT formula?

A: $f'(c) = [f(b) - f(a)] / (b - a)$ for some $c \in (a,b)$

Q5: What does LMVT represent physically?

A: At some point, instantaneous rate of change = average rate of change

Q6: LMVT vs Rolle's — relationship?

A: Rolle's is special case of LMVT when $f(a) = f(b) \to f'(c) = 0/(b-a) = 0$

Q7: CMVT formula?

A: $f'(c)/g'(c) = [f(b)-f(a)] / [g(b)-g(a)]$

Q8: Extra condition in CMVT?

A: $g'(x) \neq 0$ for all $x \in (a,b)$ — ensures denominator $\neq 0$

Q9: CMVT reduces to LMVT when?

A: When $g(x) = x$, since $g'(x)=1$ and $g(b)-g(a) = b-a$

Q10: General term of Taylor series?

A: $f^{(n)}(a) \cdot (x-a)^n / n!$ summed from $n = 0$ to ∞

Q11: Maclaurin series is Taylor series at a = ?

A: $a = 0$. General term: $f^{(n)}(0) \cdot x^n / n!$

Q12: Maclaurin expansion of e^x?

A: $e^x = 1 + x + x^2/2! + x^3/3! + \cdots$ (all powers, all positive)

Q13: Maclaurin expansion of sin x?

A: $\sin x = x - x^3/3! + x^5/5! - \cdots$ (odd powers, alternating signs)

Q14: Maclaurin expansion of cos x?

A: $\cos x = 1 - x^2/2! + x^4/4! - \cdots$ (even powers, starts with 1)

Q15: Maclaurin expansion of ln(1+x)?

A: $\ln(1+x) = x - x^2/2 + x^3/3 - \cdots$ (no factorials! valid: $-1<x\leq1$)

Q16: Convergence of ln(1+x)?

A: Valid for $-1 < x \leq 1$ only. Diverges for $|x| > 1$.

Q17: Lagrange's remainder $R_n(x)$?

A: $R_n(x) = f^{(n+1)}(c) \cdot (x-a)^{n+1} / (n+1)!$ for some c between a and x

Q18: Small angle: $\sin x \approx$?

A: $\sin x \approx x$ (first term of Maclaurin series, valid for small x in radians)

Q19: Differentiability on open or closed interval?

A: Differentiability required on OPEN interval (a, b). Continuity on $[a, b]$.

Q20: Does differentiability imply continuity?

A: YES. Differentiable $\Longrightarrow$ Continuous. But continuous $\neq$ differentiable (e.g. $|x|$)

Q21: Binomial series $(1+x)^n$ valid for?

A: $|x| < 1$ for non-integer n. For positive integer n: finite (exact) expansion.

Q22: CMVT is used to prove which rule?

A: L'Hôpital's Rule (for 0/0 and ∞/∞ indeterminate forms)

Q23: What is $\lim[x\to0] \sin(x)/x$ using series?

A: $\sin(x)/x = (x - x^3/6 + \cdots)/x = 1 - x^2/6 + \cdots \to 1$

Q24: What is $\lim[x\to0] (e^x-1)/x$ using series?

A: $(e^x-1)/x = (x + x^2/2 + \cdots)/x = 1 + x/2 + \cdots \to 1$

Q25: Series for e^{-x}?

A: $e^{-x} = 1 - x + x^2/2! - x^3/3! + \cdots$ (alternating signs)

COMMON MISTAKES — AVOID THESE IN EXAMS!

Mistake 1: Forgetting to verify ALL conditions before applying theorem
Always check: (1) Continuity on [a,b], (2) Differentiability on (a,b), and for Rolle's: (3) f(a)=f(b). If any condition fails → state theorem does NOT apply.

Mistake 2: Applying theorems to wrong intervals (open vs closed)
Continuity needed on CLOSED [a,b]. Differentiability needed on OPEN (a,b). The conclusion c must be in OPEN (a,b), not at endpoints.

Mistake 3: Confusing Maclaurin series denominators
For e^x, sin x, cos x: denominators are n! (factorials). For ln(1+x): denominators are just n (no factorial!). Mixing these up is the most common error.

Mistake 4: Wrong sign pattern in series
sin x: starts +x, $-x^3$, $+x^5$ (odd, alternating). cos x: starts +1, $-x^2$, $+x^4$ (even, alternating). ln(1+x): starts +x, $-x^2$, $+x^3$ (all, alternating).

Mistake 5: Forgetting convergence conditions
e^x, sin x, cos x converge for ALL x. But ln(1+x) converges only for $-1 < x \le 1$, and $(1+x)^n$ for $|x| < 1$. Never use outside these ranges!

Mistake 6: Forgetting g′(x) ≠ 0 in CMVT
Always explicitly verify $g'(x) \neq 0$ on (a,b). If $g'(c) = 0$ somewhere, the ratio $f'(c)/g'(c)$ is undefined and CMVT FAILS.

EXAM TIPS & SMART STRATEGIES

Tip 1 — Check conditions mechanically: For every MVT problem, write the 3-step checklist (Continuous? Differentiable? f(a)=f(b)?) before starting. Examiners give marks for this!

Tip 2 — Polynomial functions are always safe: Any polynomial is continuous and differentiable everywhere. So all 3 theorems always apply to polynomials on any [a,b].

Tip 3 — Shortcut for series expansion accuracy: For approximations to 4 decimal places, keep terms until $x^n/n! < 0.00001$. Usually 4–5 terms are sufficient.

Tip 4 — Use sin x ≈ x for small angles: When x < 0.1 radians, sin x ≈ x with less than 0.17% error. This is sufficient for most engineering approximation problems.

Tip 5 — Remember the hierarchy: CMVT is the most general → LMVT is a special case → Rolle's is a further special case. In exams, if asked to "compare", draw the hierarchy.

Tip 6 — For Maclaurin problems: Always start by finding f(0), f′(0), f″(0), f‴(0). If the pattern is not obvious after 4 derivatives, write out all terms explicitly rather than guessing.

SHORTCUT TRICKS

Shortcut	Description
Derive cos from sin	d/dx(sin x) = cos x → differentiate sin x series term-by-term to get cos x series
Get ln(1−x) instantly	Replace x with −x in ln(1+x): result is $-x - x^2/2 - x^3/3 - \cdots$
Rolle's converse check	If f(a) ≠ f(b), immediately say Rolle's does not apply (no need to check other conditions)
LMVT "c" range check	After solving for c, verify a < c < b. If c is outside, there is a calculation error.
sinh/cosh from e^x	$\sinh x = (e^x - e^{-x})/2$ → odd terms only; $\cosh x = (e^x + e^{-x})/2$ → even terms only

ROLLE'S THEOREM

$$f'(c) = 0$$

Conditions: cont.[a,b] + diff.(a,b) + f(a)=f(b)

LMVT (LAGRANGE)

$$f'(c) = [f(b) - f(a)] / (b - a)$$

Conditions: cont.[a,b] + diff.(a,b)

CMVT (CAUCHY)

$$f'(c)/g'(c) = [f(b)-f(a)] / [g(b)-g(a)]$$

Extra: $g'(x) \neq 0$ on (a,b)

TAYLOR SERIES

$$f(x) = f(a) + (x-a)f'(a) + (x-a)^2 f''(a)/2! + (x-a)^3 f'''(a)/3! + \cdots + R_n(x)$$

$R_n(x) = f^{(n+1)}(c)\cdot(x-a)^{n+1}/(n+1)!$

MACLAURIN SERIES — STANDARD EXPANSIONS

$$e^x = 1 + x + x^2/2! + x^3/3! + x^4/4! + \cdots$$

$$\sin x = x - x^3/3! + x^5/5! - x^7/7! + \cdots$$

$$\cos x = 1 - x^2/2! + x^4/4! - x^6/6! + \cdots$$

$$\ln(1+x) = x - x^2/2 + x^3/3 - x^4/4 + \cdots$$

$$(1+x)^n = 1 + nx + n(n-1)x^2/2! + n(n-1)(n-2)x^3/3! + \cdots$$

DERIVED SERIES

$e^{-x} = 1-x+x^2/2!-x^3/3!+\cdots$

$\ln(1-x) = -x-x^2/2-x^3/3-\cdots$

$\sinh x = x+x^3/3!+x^5/5!+\cdots$

$\cosh x = 1+x^2/2!+x^4/4!+\cdots$

CONVERGENCE RANGES

e^x : all $x \in \mathbb{R}$

$\sin x$: all $x \in \mathbb{R}$

$\cos x$: all $x \in \mathbb{R}$

$\ln(1+x)$: $-1 < x \leq 1$

$(1+x)^n$: $|x| < 1$

SMALL ANGLE (X IN RAD)

$\sin x \approx x$

$\cos x \approx 1 - x^2/2$

$\tan x \approx x$

$e^x \approx 1 + x$ ($|x| \ll 1$)

$\ln(1+x) \approx x$ ($|x| \ll 1$)

🏆 THEOREM HIERARCHY: CMVT ⊃ LMVT ⊃ ROLLE'S THEOREM

Rolle's Special Case
Set f(a) = f(b) in LMVT
→ f′(c) = 0/(b−a) = 0 ✓

LMVT Special Case
Set g(x) = x in CMVT
→ g′(x) = 1, g(b)−g(a) = b−a ✓

CMVT Application
Proves L'Hôpital's Rule
for 0/0 and ∞/∞ forms ✓